Sphygmomanometer



Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 12 labs

Covers these lab items: Blood Pressure Measuring (Sphygmomanometer)- Mercury Free; Blood Pressure Monitor; Blood pressure monitor (manual and/or digital); BP Machine; B.P. Monitoring Machine; Digital BP machine; Electronic Blood Pressure Monitoring Machine; Sphygmomanometer

What it is used for

A sphygmomanometer measures blood pressure using a cuff and a manual or digital measurement system. Correct cuff placement, patient preparation, and careful reading help produce more reliable results.

Main parts

- Cuff wraps around the upper arm or another approved measurement site.
- Inflation bulb or pump inflates the cuff on manual models.
- Valve releases cuff pressure gradually on manual models.
- Pressure gauge displays pressure on aneroid or mercury-free manual models.
- Digital display shows blood pressure and pulse on electronic models.
- Tubing connects the cuff to the pump or measuring unit.
- Power button and batteries operate many digital models.

Before you start

1. Use the device under teacher or healthcare professional supervision and follow its instructions.
2. Explain the procedure and allow the person to sit quietly for about five minutes.
3. Ask the person to avoid talking and keep feet flat with the back supported.
4. Use a correctly sized cuff on a bare upper arm unless the device specifies otherwise.
5. Check the cuff, tubing, connectors, display, and battery condition for damage.
6. Avoid measuring over clothing or immediately after exercise, caffeine, or smoking when possible.

How to use

1. Position the person's arm supported at heart level with the palm facing upward.
2. Wrap the cuff snugly around the bare upper arm according to the cuff markings.
3. For a manual device, place the stethoscope over the brachial artery if trained to do so.
4. Inflate the cuff according to the device instructions and the trained measurement method.
5. Release pressure slowly on manual models while listening for Korotkoff sounds if trained.
6. For digital models, press the start button and keep the person still during measurement.
7. Wait for the reading to appear and note systolic pressure, diastolic pressure, and pulse if displayed.
8. Record the result with the time and relevant observations according to school procedure.

9. If a reading seems unusual, allow a short rest and repeat the measurement correctly.
10. Remove the cuff, switch off the device, and clean reusable surfaces as instructed.

⚠ Safety

- Do not use a cuff on an injured arm or an arm with a medical restriction.
- Keep the cuff and tubing clean and avoid excessive tightening or prolonged inflation.
- Never diagnose a medical condition from one reading; refer concerning results to a healthcare professional.
- Stop if the person experiences pain, dizziness, or distress during measurement.
- Do not share cuffs without cleaning them according to infection-control instructions.
- Keep mercury-free equipment intact and dispose of damaged electronic parts and batteries appropriately.

Care & storage

- Clean the cuff and device using only manufacturer-approved cleaning methods.
- Store the device in a dry protective case away from heat and moisture.
- Check batteries, tubing, cuff fabric, and connectors regularly for damage.
- Arrange calibration or accuracy checks at the intervals recommended by the manufacturer or healthcare facility.

Common mistakes to avoid

- Measuring over clothing or using an incorrectly sized cuff.
- Allowing the person to talk, move, or keep the arm unsupported.
- Placing the arm below or above heart level during measurement.
- Recording the reading incorrectly or interpreting one result as a diagnosis.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.